THIRD EDITION

Introduction to Random Signals and Applied Kalman Filtering

WITH MATLAB EXERCISES AND SOLUTIONS

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1. Convert the scalar equation, Eq. (5.4.10), to vector form using the "phase variable" method given in Section 5.3. The result is given by Eqs. (5.4.11) and (5.4.12).
2. A white Gaussian sequence with unity variance is needed for the simulation. This is easily obtained using the MATLAB random number generator.
3. An initial condition on $\mathbf{x}(k)$ must be specified. In this example it is a 2-tuple. The choice of initial condition will govern the initial transient in much the same way as in the continuous-time problem.
4. Programming the recursive equation for $\mathbf{x}(k)$ is routine, once the initial vector $\mathbf{x}(0)$ is specified. Program Eq. (5.4.11).
5. Finally, the original $y(k)$ process is reconstructed from $\mathbf{x}(k)$ at each step using the output equation, Eq. (5.4.12).

In summary, simulation is an important tool in analyzing systems with random inputs. With modern computer technology, simulation is relatively easy to do, and it is often the best way to get answers to otherwise intractable problems.

5.5 THE DISCRETE KALMAN FILTER

R. E. Kalman's paper describing a recursive solution of the discrete-data linear filtering problem was published in 1960 (1). About this same time, advances in digital computer technology made it possible to consider implementing his recursive solution in a number of real-time applications. This was a fortuitous circumstance, and Kalman filtering caught hold almost immediately. We will consider some examples shortly, but we must first develop the Kalman filter recursive equations, which are, in effect, the "filter."

We begin by assuming the random process to be estimated can be modeled in the form

$$\mathbf{x}_{k+1} = \boldsymbol{\phi}_k \mathbf{x}_k + \mathbf{w}_k \tag{5.5.1}$$

The observation (measurement) of the process is assumed to occur at discrete points in time in accordance with the linear relationship

$$\mathbf{z}_k = \mathbf{H}_k \mathbf{x}_k + \mathbf{v}_k \tag{5.5.2}$$

Some elaboration on notation and the various terms of Eqs. (5.5.1) and (5.5.2) is in order:

$\mathbf{x}_k$ = $(n \times 1)$ process state vector at time t_k

$\boldsymbol{\phi}_k$ = $(n \times n)$ matrix relating $\mathbf{x}_k$ to $\mathbf{x}_{k+1}$ in the absence of a forcing function (if $\mathbf{x}_k$ is a sample of continuous process, $\boldsymbol{\phi}_k$ is the usual state transition matrix)

$\mathbf{w}_k$ = $(n \times 1)$ vector—assumed to be a white sequence with known covariance structure

$\mathbf{z}_k$ = $(m \times 1)$ vector measurement at time t_k

$\mathbf{H}_k$ = $(m \times n)$ matrix giving the ideal (noiseless) connection between the measurement and the state vector at time t_k

$\mathbf{v}_k$ = $(m \times 1)$ measurement error—assumed to be a white sequence with known covariance structure and having zero crosscorrelation with the $\mathbf{w}_k$ sequence

The covariance matrices for the $\mathbf{w}_k$ and $\mathbf{v}_k$ vectors are given by

$$E[\mathbf{w}_k\mathbf{w}_i^T] = \begin{cases} \mathbf{Q}_k, & i = k \\ 0, & i \neq k \end{cases} \tag{5.5.3}$$

$$E[\mathbf{v}_k\mathbf{v}_i^T] = \begin{cases} \mathbf{R}_k, & i = k \\ 0, & i \neq k \end{cases} \tag{5.5.4}$$

$$E[\mathbf{w}_k\mathbf{v}_i^T] = 0, \qquad \text{for all } k \text{ and } i \tag{5.5.5}$$

We assume at this point that we have an initial estimate of the process at some point in time t_k, and that this estimate is based on all our knowledge about the process prior to t_k. This prior (or *a priori*) estimate will be denoted as $\hat{\mathbf{x}}_k^-$ where the "hat" denotes estimate, and the "super minus" is a reminder that this is our best estimate prior to assimilating the measurement at t_k. (Note that super minus as used here is not related in any way to the super minus notation used in spectral factorization.) We also assume that we know the error covariance matrix associated with $\hat{\mathbf{x}}_k^-$. That is, we define the estimation error to be

$$\mathbf{e}_k^- = \mathbf{x}_k - \hat{\mathbf{x}}_k^- \tag{5.5.6}$$

and the associated error covariance matrix is*

$$\mathbf{P}_k^- = E[\mathbf{e}_k^-\mathbf{e}_k^{-T}] = E[(\mathbf{x}_k - \hat{\mathbf{x}}_k^-)(\mathbf{x}_k - \hat{\mathbf{x}}_k^-)^T] \tag{5.5.7}$$

* We tacitly assume here that the estimation error has zero mean, and thus it is proper to refer to $E[\mathbf{e}_k^-\mathbf{e}_k^{-T}]$ as a covariance matrix. It is also, of course, a moment matrix, but it is usually not referred to as such.

In many cases, we begin the estimation problem with no prior measurements. Thus, in this case, if the process mean is zero, the initial estimate is zero, and the associated error covariance matrix is just the covariance matrix of $\mathbf{x}$ itself.

With the assumption of a prior estimate $\hat{\mathbf{x}}_k^-$, we now seek to use the measurement $\mathbf{z}_k$ to improve the prior estimate. We choose a linear blending of the noisy measurement and the prior estimate in accordance with the equation

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k(\mathbf{z}_k - \mathbf{H}_k\hat{\mathbf{x}}_k^-) \tag{5.5.8}$$

where

$$\hat{\mathbf{x}}_k = \text{updated estimate}$$
$$\mathbf{K}_k = \text{blending factor} \quad \text{(yet to be determined)}$$

The justification of the special form of Eq. (5.5.8) will be deferred until Section 5.8. The problem now is to find the particular blending factor $\mathbf{K}_k$ that yields an updated estimate that is optimal in some sense. Just as in the Wiener solution, we use minimum mean-square error as the performance criterion. Toward this end, we first form the expression for the error covariance matrix associated with the updated (*a posteriori*) estimate.

$$\mathbf{P}_k = E[\mathbf{e}_k\mathbf{e}_k^T] = E[(\mathbf{x}_k - \hat{\mathbf{x}}_k)(\mathbf{x}_k - \hat{\mathbf{x}}_k)^T] \tag{5.5.9}$$

Next, we substitute Eq. (5.5.2) into Eq. (5.5.8) and then substitute the resulting expression for $\hat{\mathbf{x}}_k$ into Eq. (5.5.9). The result is

$$\begin{aligned}\mathbf{P}_k = E\{&[(\mathbf{x}_k - \hat{\mathbf{x}}_k^-) - \mathbf{K}_k(\mathbf{H}_k\mathbf{x}_k + \mathbf{v}_k - \mathbf{H}_k\hat{\mathbf{x}}_k^-)] \\ &[(\mathbf{x}_k - \hat{\mathbf{x}}_k^-) - \mathbf{K}_k(\mathbf{H}_k\mathbf{x}_k + \mathbf{v}_k - \mathbf{H}_k\hat{\mathbf{x}}_k^-)]^T\}\end{aligned} \tag{5.5.10}$$

Now, performing the indicated expectation and noting the $(\mathbf{x}_k - \hat{\mathbf{x}}_k^-)$ is the a priori estimation error that is uncorrelated with the measurement error $\mathbf{v}_k$, we have

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^-(\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)^T + \mathbf{K}_k\mathbf{R}_k\mathbf{K}_k^T \tag{5.5.11}$$

Notice here that Eq. (5.5.11) is a perfectly general expression for the updated error covariance matrix, and it applies for any gain $\mathbf{K}_k$, suboptimal or otherwise.

Returning to the optimization problem, we wish to find the particular $\mathbf{K}_k$ that minimizes the individual terms along the major diagonal of $\mathbf{P}_k$, because these terms represent the estimation error variances for the elements of the state vector being estimated. The optimization can be done in a number of ways. We will do this using a straightforward differential calculus approach, and to do so we need two matrix differentiation formulas. They are

$$\frac{d[\text{trace}(\mathbf{AB})]}{d\mathbf{A}} = \mathbf{B}^T \quad (\mathbf{AB} \text{ must be square}) \tag{5.5.12}$$

$$\frac{d[\text{trace}(\mathbf{ACA}^T)]}{d\mathbf{A}} = 2\mathbf{AC} \quad (\mathbf{C} \text{ must be symmetric}) \tag{5.5.13}$$

where the derivative of a scalar with respect to a matrix is defined as

$$\frac{ds}{d\mathbf{A}} = \begin{bmatrix} \dfrac{ds}{da_{11}} & \dfrac{ds}{da_{12}} & \cdots \\ \dfrac{ds}{da_{21}} & \dfrac{ds}{da_{22}} & \cdots \\ \vdots & & \end{bmatrix} \tag{5.5.14}$$

Proof of these two differentiation formulas will be left as an exercise (see Problem 5.16). We will now expand the general form for $\mathbf{P}_k$, Eq. (5.5.11), and rewrite it in the form:

$$\mathbf{P}_k = \mathbf{P}_k^- - \mathbf{K}_k\mathbf{H}_k\mathbf{P}_k^- - \mathbf{P}_k^-\mathbf{H}_k^T\mathbf{K}_k^T + \mathbf{K}_k(\mathbf{H}_k\mathbf{P}_k^-\mathbf{H}_k^T + \mathbf{R}_k)\mathbf{K}_k^T \tag{5.5.15}$$

Notice that the second and third terms are linear in $\mathbf{K}_k$ and that the fourth term is quadratic in $\mathbf{K}$. The two matrix differentiation formulas may now be applied to Eq. (5.5.15). We wish to minimize the trace of $\mathbf{P}$ because it is the sum of the mean-square errors in the estimates of all the elements of the state vector. We can use the argument here that the individual mean-square errors are also minimized when the total is minimized, provided that we have enough degrees of freedom in the variation of $\mathbf{K}_k$, which we do in this case. We proceed now to differentiate the trace of $\mathbf{P}_k$ with respect to $\mathbf{K}_k$, and we note that the trace of $\mathbf{P}_k^-\mathbf{H}_k^T\mathbf{K}_k^T$ is equal to the trace of its transpose $\mathbf{K}_k\mathbf{H}_k\mathbf{P}_k^-$. The result is

$$\frac{d(\text{trace } \mathbf{P}_k)}{d\mathbf{K}_k} = -2(\mathbf{H}_k\mathbf{P}_k^-)^T + 2\mathbf{K}_k(\mathbf{H}_k\mathbf{P}_k^-\mathbf{H}_k^T + \mathbf{R}_k) \tag{5.5.16}$$

We now set the derivative equal to zero and solve for the optimal gain. The result is

$$\mathbf{K}_k = \mathbf{P}_k^-\mathbf{H}_k^T(\mathbf{H}_k\mathbf{P}_k^-\mathbf{H}_k^T + \mathbf{R}_k)^{-1} \tag{5.5.17}$$

This particular $\mathbf{K}_k$, namely, the one that minimizes the mean-square estimation error, is called the *Kalman gain.*

The covariance matrix associated with the optimal estimate may now be computed. Referring to Eq. (5.5.11), we have

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^-(\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)^T + \mathbf{K}_k\mathbf{R}_k\mathbf{K}_k^T \tag{5.5.18}$$

$$= \mathbf{P}_k^- - \mathbf{K}_k\mathbf{H}_k\mathbf{P}_k^- - \mathbf{P}_k^-\mathbf{H}_k^T\mathbf{K}_k^T + \mathbf{K}_k(\mathbf{H}_k\mathbf{P}_k^-\mathbf{H}_k^T + \mathbf{R}_k)\mathbf{K}_k^T \tag{5.5.19}$$

Routine substitution of the optimal gain expression, Eq. (5.5.17), into Eq. (5.5.19) leads to

$$\mathbf{P}_k = \mathbf{P}_k^- - \mathbf{P}_k^-\mathbf{H}_k^T(\mathbf{H}_k\mathbf{P}_k^-\mathbf{H}_k^T + \mathbf{R}_k)^{-1}\mathbf{H}_k\mathbf{P}_k^- \tag{5.5.20}$$

or

$$\mathbf{P}_k = \mathbf{P}_k^- - \mathbf{K}_k(\mathbf{H}_k\mathbf{P}_k^-\mathbf{H}_k^T + \mathbf{R}_k)\mathbf{K}_k^T \tag{5.5.21}$$

or

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^- \tag{5.5.22}$$

Note that we have four expressions for computing the updated $\mathbf{P}_k$ from the prior $\mathbf{P}_k^-$. Three of these, Eqs. (5.5.20), (5.5.21), and (5.5.22), are only valid for the optimal gain condition. However, Eq. (5.5.18) is valid for any gain, optimal or suboptimal. All four equations yield identical results for optimal gain with perfect arithmetic. We note, though, that in the real engineering world Kalman filtering is a numerical procedure, and some of the *P*-update equations may perform better numerically than others under unusual conditions. More will be said of this later in Chapter 6. For now, we will list the simplest update equation, that is, Eq. (5.5.22), as the usual way to update the error covariance. One should remember, though, that there are alternative equations for implementing the error covariance update.

We now have a means of assimilating the measurement at t_k by the use of Eq. (5.5.8) with $\mathbf{K}_k$ set equal to the Kalman gain as given by Eq. (5.5.17). Note that we need $\hat{\mathbf{x}}_k^-$ and $\mathbf{P}_k^-$ to accomplish this, and we can anticipate a similar need at the next step in order to make optimal use of the measurement $\mathbf{z}_{k+1}$. The updated estimated $\hat{\mathbf{x}}_k$ is easily projected ahead via the transition matrix. We are justified in ignoring the contribution of $\mathbf{w}_k$ in Eq. (5.5.1) because it has zero mean and is not correlated with any of the previous $\mathbf{w}$'s.* Thus, we have

$$\hat{\mathbf{x}}_{k+1}^- = \boldsymbol{\phi}_k\hat{\mathbf{x}}_k \tag{5.5.23}$$

* Recall that in our notation $\mathbf{w}_k$ is the process noise that accumulates during the step ahead from t_k to t_{k+1}. This is purely a matter of notation (but an important one), and in some books it is denoted as $\mathbf{w}_{k+1}$ rather than $\mathbf{w}_k$ (6, 9). Consistency in notation is the important thing here. Conceptually, we are thinking of doing real-time filtering in contrast to smoothing, which we usually think of doing offline (see Chapter 8). Therefore, if we begin with a discrete ARMA model, the white forcing sequence $w(k)$ must conform to this same step-ahead notation, and we need to restrict the order of the MA part of the model to be at least one less than the AR part (see Section 5.3). This then assures us that $w(k)$ contributes only to the state vector *after* time t_k and not before.

The error covariance matrix associated with $\hat{\mathbf{x}}_{k+1}^-$ is obtained by first forming the expression for the a priori error

$$\begin{aligned}\mathbf{e}_{k+1}^- &= \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1}^- \\ &= (\boldsymbol{\phi}_k \mathbf{x}_k + \mathbf{w}_k) - \boldsymbol{\phi}_k \hat{\mathbf{x}}_k \\ &= \boldsymbol{\phi}_k \mathbf{e}_k + \mathbf{w}_k \end{aligned} \tag{5.5.24}$$

We now note that $\mathbf{w}_k$ and $\mathbf{e}_k$ have zero crosscorrelation, because $\mathbf{w}_k$ is the process noise for the step ahead of t_k. Thus, we can write the expression for $\mathbf{P}_{k+1}^-$ as

$$\begin{aligned}\mathbf{P}_{k+1}^- &= E[\mathbf{e}_{k+1}^- \mathbf{e}_{k+1}^{-T}] = E[(\boldsymbol{\phi}_k \mathbf{e}_k + \mathbf{w}_k)(\boldsymbol{\phi}_k \mathbf{e}_k + \mathbf{w}_k)^T] \\ &= \boldsymbol{\phi}_k \mathbf{P}_k \boldsymbol{\phi}_k^T + \mathbf{Q}_k \end{aligned} \tag{5.5.25}$$

We now have the needed quantities at time t_{k+1}, and the measurement $\mathbf{z}_{k+1}$ can be assimilated just as in the previous step.

Equations (5.5.8), (5.5.17), (5.5.22), (5.5.23), and (5.5.25) comprise the Kalman filter recursive equations. It should be clear that once the loop is entered, it can be continued ad infinitum. The pertinent equations and the sequence of computational steps are shown pictorially in Fig. 5.8. This summarizes what is now known as the *Kalman filter*.

Before we proceed to some examples, it is interesting to reflect on the Kalman filter in perspective. If you were to stumble onto the recursive process of Fig. 5.8 without benefit of previous history, you might logically ask, "Why in the world did somebody call that a filter? It looks more like a computer algorithm." You would, of course, be quite right in your observation. The Kalman filter is just a computer algorithm for processing discrete measurements (the input) into optimal estimates (the output). Its roots, though, go back to the

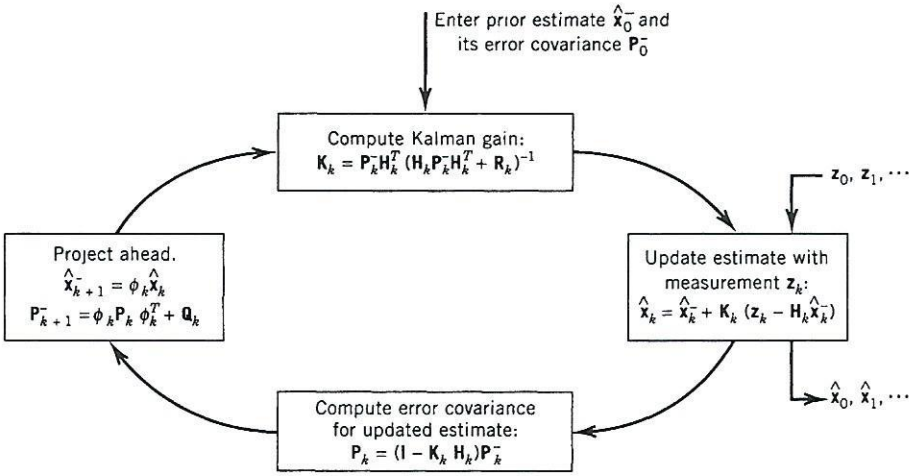


Figure 5.8 Kalman filter loop.

days when filters were made of electrical elements wired together in such a way as to yield the desired frequency response. The design was often heuristic. Wiener then came on the scene in the 1940s and added a more sophisticated type of filter problem. The end result of his solution was a filter weighting function or a corresponding transfer function in the complex domain. Implementation in terms of electrical elements was left as a further exercise for the designer. The discrete-time version of the Wiener problem remained unsolved (in a practical sense, at least) until Kalman's paper of 1960. Even though his presentation appeared to be quite abstract at first glance, engineers soon realized that this work provided a practical solution to a number of unsolved filtering problems, especially in the field of navigation. More than 35 years have elapsed since Kalman's original paper, and there are still numerous current papers dealing with new applications and variations on the basic Kalman filter. It has withstood the test of time!

5.6 SCALAR KALMAN FILTER EXAMPLES

The basic recursive equations for the Kalman filter were presented in Section 5.5. Two simple scalar examples will illustrate the use of these equations.

EXAMPLE 5.7

Wiener (Brownian Motion) Process Consider the Wiener process described in Fig. 5.9. Recall that this process has Gaussian statistics and is assumed to be zero at $t = 0$. Assume that we have a sequence of independent noisy measurements taken at unit intervals as shown in Fig. 5.9 and let the standard deviation of the measurement error be one-half. We wish to determine the optimal estimate of the process at the sample times 0, 1, 2, 3, . . . , etc.

The model parameters for the Kalman filter may be computed as follows: The process differential equation for this case is

$$\dot{x} = u(t) \tag{5.6.1}$$

Therefore, the transition matrix is

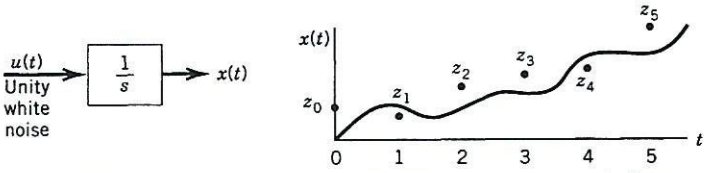


Figure 5.9 Block diagram of Wiener process and typical sample function of the process.

$$\phi_k = 1 \tag{5.6.2}$$

The Q_k matrix is computed as

$$\begin{aligned} Q_k = E[w_k, w_k] &= E\left[\int_0^1 1 \cdot u(\xi)\, d\xi \int_0^1 1 \cdot u(\eta)\, d\eta\right] \\ &= \int_0^1 \int_0^1 E[u(\xi)u(\eta)]\, d\xi\, d\eta = \int_0^1 \int_0^1 \delta(\xi - \eta)\, d\xi\, d\eta = 1 \end{aligned} \tag{5.6.3}$$

Since the measurement has a direct one-to-one correspondence to the process $x(t)$, the H_k matrix is

$$H_k = 1 \tag{5.6.4}$$

The variance of the measurement error is

$$R_k = (\text{Standard deviation})^2 = (\tfrac{1}{2})^2 = \tfrac{1}{4} \tag{5.6.5}$$

Our initial estimate at $t = 0$ is $\hat{x}_0^-$ and is zero because the Wiener process is defined to begin at zero. Furthermore, because of this prior knowledge about the process, the error associated with our initial estimate is zero; that is, the a priori estimate, $\hat{x}_0^- = 0$, is perfect by definition of the process. Thus,

$$P_0^- = 0 \tag{5.6.6}$$

We now have all the parameters needed for the Kalman filter and are ready to begin the recursive process. Referring to Fig. 5.8, we enter the loop at $t = 0$ and process the first measurement.

Step 1: $t = 0$ (Subscripts will be dropped for constant parameters.)
Compute gain:

$$\begin{aligned} K_0 &= P_0^- H^T (H P_0^- H^T + R)^{-1} \\ &= \frac{0}{.25} = 0 \end{aligned} \tag{5.6.7}$$

Update estimate:

$$\begin{aligned} \hat{x}_0 &= \hat{x}_0^- + K_0(z_0 - H\hat{x}_0^-) \\ &= \hat{x}_0^- = 0 \end{aligned} \tag{5.6.8}$$

Update P:

$$P_0 = (I - K_0H)P_0^-$$
$$= 1 \cdot P_0^- = 0 \tag{5.6.9}$$

Project ahead:

$$\hat{x}_1^- = \phi\hat{x}_0 = 1 \cdot 0 = 0$$
$$P_1^- = \phi P_0\phi^T + Q \tag{5.6.10}$$
$$= 1 \cdot 0 \cdot 1 + 1 = 1 \tag{5.6.11}$$

Note that the measurement at $t = 0$ is given zero weight relative to the prior estimate. This makes good sense because the initial estimate is known to be perfect, whereas the measurement is known to be noisy.

Step 2: $t = 1$
Compute gain:

$$K_1 = P_1^-H^T(HP_1^-H^T + R)^{-1}$$
$$= 1 \cdot 1(1 \cdot 1 \cdot 1 + \tfrac{1}{4})^{-1} = \tfrac{4}{5} \tag{5.6.12}$$

Update estimate:

$$\hat{x}_1 = \hat{x}_1^- + K_1(z_1 - H\hat{x}_1^-)$$
$$= 0 + \tfrac{4}{5}(z_1) \tag{5.6.13}$$

Update P:

$$P_1 = (I - K_1H)P_1^-$$
$$= (1 - \tfrac{4}{5} \cdot 1)1 = \tfrac{1}{5} \tag{5.6.14}$$

Project ahead:

$$\hat{x}_2^- = \phi\hat{x}_1 = 1 \cdot \hat{x}_1$$
$$P_2^- = \phi P_1\phi_T + Q \tag{5.6.15}$$
$$= 1 \cdot \tfrac{1}{5} \cdot 1 + 1 = \tfrac{6}{5} \tag{5.6.16}$$

This could now be carried on ad infinitum. Note that with step 2 the filter begins to give the measurement some weight in determining the optimal estimate. As times goes on, the filter depends more and more on the measurements and less on the initial assumptions. ■

EXAMPLE 5.8

Scalar Gauss–Markov Process Consider a stationary Gauss–Markov process whose autocorrelation function is

$$R_x(\tau) = 1 \cdot e^{-|\tau|} \tag{5.6.17}$$

Clearly, the correlation time and variance for this process are both unity.

Therefore, the spectral function for the process is

$$S_x(s) = \frac{2}{-s^2 + 1} = \frac{\sqrt{2}}{s + 1} \cdot \frac{\sqrt{2}}{-s + 1} \tag{5.6.18}$$

and the shaping filter that shapes white noise into the process is shown in Fig. 5.10. Thus, the state equation for this process is

$$\dot{x} = -x + \sqrt{2}u(t) \tag{5.6.19}$$

Suppose we have a sequence of noisy measurements of this process taken .02 sec apart beginning at $t = 0$. The measurement error will be assumed to have a variance of unity. We wish to process these via a Kalman filter and obtain an optimal estimate of $x(t)$. First, we need to determine the filter parameters ϕ_k, H_k, Q_k, and R_k.

The state transition matrix (scalar in this case) is

$$\phi_k = e^{-.02} \approx .9802 \tag{5.6.20}$$

The measurement relationship to x is

$$H_k = 1 \tag{5.6.21}$$

The input noise sequence is

$$\begin{aligned} Q_k = E[w_k^2] &= E\left[\int_0^{.02} \sqrt{2}e^{-\xi}u(\xi)\,d\xi \int_0^{.02} \sqrt{2}e^{-\eta}u(\eta)\,d\eta\right] \\ &= \int_0^{.02} (\sqrt{2}e^{-\eta})^2\,d\eta = 1 - e^{-2(.02)} \approx .03921 \end{aligned} \tag{5.6.22}$$

The measurement error variance is

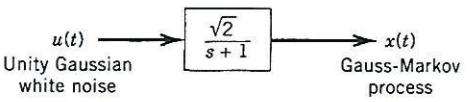


Figure 5.10 Shaping filter for Gauss–Markov process.

$$R_k = 1 \tag{5.6.23}$$

We also need the initial conditions $\hat{x}_0^-$ and P_0^- to enter the recursive loop. In this case, we have assumed that the process is Gauss–Markov and stationary with a known autocorrelation function. We have no measurements prior to $t = 0$, but the assumed knowledge of the process autocorrelation function tells us the process has zero mean and a variance of unity. This is important information and enables us to start the recursive process at $t = 0$ with initial conditions

$$\hat{x}_0^- = 0 \tag{5.6.24}$$

$$P_0^- = 1 \tag{5.6.25}$$

Now that the four filter parameters and initial conditions are determined, it is a routine matter to cycle through the Kalman filter loop shown in Fig. 5.8 as many times as desired.

In order to add a note of realism to this example, the Markov process and noisy measurement situation just described were simulated using the methods described in Section 5.4. The results for the first 51 steps of the simulation are shown in Fig. 5.11. The discrete measurements z_k are shown as triangles, and it should be obvious that we have postulated a very noisy measurement situation for this example. In spite of this, though, the filter does a reasonably good job of tracking $x(t)$. After about the first 20 steps, the filter settles down to a steady-state condition where the Kalman gain is about .165 and the standard deviation of the filter error is about .4. That is, $\sqrt{P_k}$ in the steady state is about .4 units.

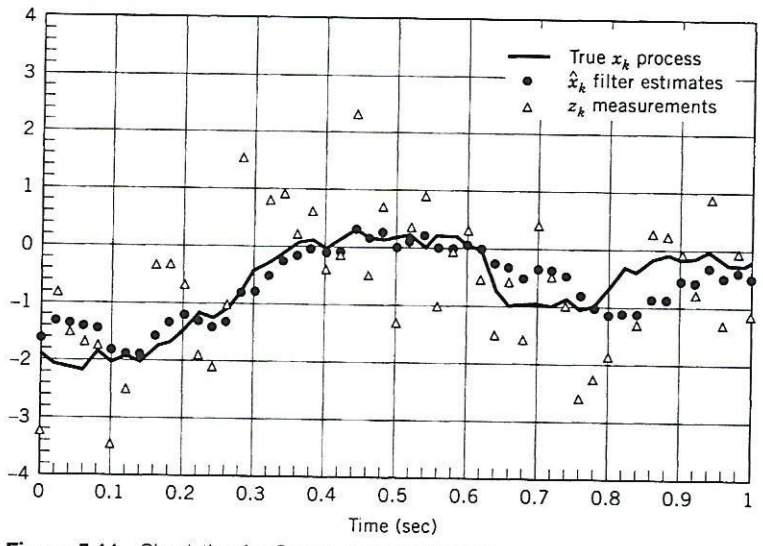


Figure 5.11 Simulation for Gauss–Markov example.

9

Linearization and Additional Intermediate-Level Topics on Applied Kalman Filtering

Kalman's papers of the early 1960s (1, 2) were recognized almost immediately as new and important contributions to least-squares filtering. As a result, there was a renewal of research interest in this area, and a flurry of papers expanding on Kalman's original work followed during the next decade or so. Kailath (3) gives an especially comprehensive bibliography of papers for this period. Research work in this area still continues (although perhaps at a somewhat reduced rate), and new applications and extensions continue to appear regularly in the technical literature. A few of the more significant extensions and related topics have been selected for comment here. The list is by no means comprehensive. However, there is a hierarchy of importance, and it is the authors' recommendation that the reader begin with the first topic on linearization. This is, by far, the most important section in this chapter. The other sections may be studied in any desired order as time permits.

9.1 LINEARIZATION

Some of the most successful applications of Kalman filtering have been in situations with nonlinear dynamics and/or nonlinear measurement relationships. We now examine two basic ways of linearizing the problem. One is to linearize about some nominal trajectory in state space that does not depend on the measurement data. The resulting filter is usually referred to as simply a *linearized Kalman filter*. The other method is to linearize about a trajectory that is continually updated with the state estimates resulting from the measurements. When this is done, the filter is called an *extended Kalman filter*. A brief discussion of each will now be presented.

Linearized Kalman Filter

We begin by assuming the process to be estimated and the associated measurement relationship may be written in the form

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}_d, t) + \mathbf{u}(t) \tag{9.1.1}$$

$$\mathbf{z} = \mathbf{h}(\mathbf{x}, t) + \mathbf{v}(t) \tag{9.1.2}$$

where $\mathbf{f}$ and $\mathbf{h}$ are known functions, $\mathbf{u}_d$ is a deterministic forcing function, and $\mathbf{u}$ and $\mathbf{v}$ are white-noise processes with zero crosscorrelation as before. Note that nonlinearity may enter into the problem either in the dynamics of the process or in the measurement relationship. Also, note that the forms of Eqs. (9.1.1) and (9.1.2) are somewhat restrictive in that $\mathbf{u}$ and $\mathbf{v}$ are assumed to be separate additive terms and are not included with the $\mathbf{f}$ and $\mathbf{h}$ terms. However, to do otherwise complicates the problem considerably, and thus we will stay with these restrictive forms.

Let us now assume that an approximate trajectory $\mathbf{x}^*(t)$ may be determined by some means. This will be referred to as the nominal or reference trajectory, and it is illustrated along with the actual trajectory in Fig. 9.1. The actual trajectory $\mathbf{x}(t)$ may then be written as

$$\mathbf{x}(t) = \mathbf{x}^*(t) + \Delta\mathbf{x}(t) \tag{9.1.3}$$

Equations (9.1.1) and (9.1.2) then become

$$\dot{\mathbf{x}}^* + \Delta\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}^* + \Delta\mathbf{x}, \mathbf{u}_d, t) + \mathbf{u}(t) \tag{9.1.4}$$

$$\mathbf{z} = \mathbf{h}(\mathbf{x}^* + \Delta\mathbf{x}, t) + \mathbf{v}(t) \tag{9.1.5}$$

We now assume $\Delta\mathbf{x}$ is small and approximate the $\mathbf{f}$ and $\mathbf{h}$ functions with Taylor's series expansions, retaining only first-order terms. The result is

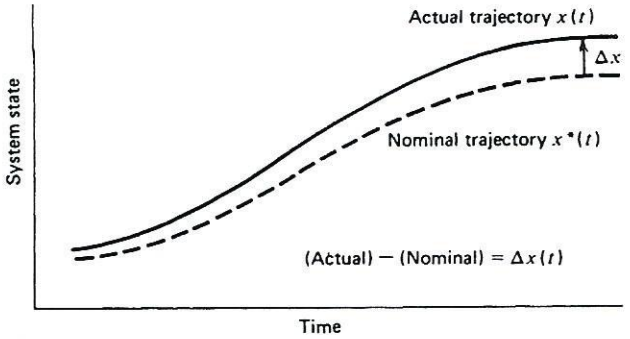


Figure 9.1 Nominal and actual trajectories for a linearized Kalman filter.

$$\dot{\mathbf{x}}^* + \Delta\dot{\mathbf{x}} \approx \mathbf{f}(\mathbf{x}^*, \mathbf{u}_d, t) + \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right]_{\mathbf{x}=\mathbf{x}^*} \cdot \Delta\mathbf{x} + \mathbf{u}(t) \tag{9.1.6}$$

$$\mathbf{z} \approx \mathbf{h}(\mathbf{x}^*, t) + \left[\frac{\partial \mathbf{h}}{\partial \mathbf{x}}\right]_{\mathbf{x}=\mathbf{x}^*} \cdot \Delta\mathbf{x} + \mathbf{v}(t) \tag{9.1.7}$$

where

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} \frac{\partial f_1}{\partial x_2} \cdots \\ \frac{\partial f_2}{\partial x_1} \frac{\partial f_2}{\partial x_2} \cdots \\ \vdots \end{bmatrix}; \qquad \frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial h_1}{\partial x_1} \frac{\partial h_1}{\partial x_2} \cdots \\ \frac{\partial h_2}{\partial x_1} \frac{\partial h_2}{\partial x_2} \cdots \\ \vdots \end{bmatrix} \tag{9.1.8}$$

It is customary to choose the nominal trajectory $\mathbf{x}^*(t)$ to satisfy the deterministic differential equation

$$\dot{\mathbf{x}}^* = \mathbf{f}(\mathbf{x}^*, \mathbf{u}_d, t) \tag{9.1.9}$$

Substituting this into (9.1.6) then leads to the linearized model

$$\Delta\dot{\mathbf{x}} = \left[\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right]_{\mathbf{x}=\mathbf{x}^*} \cdot \Delta\mathbf{x} + \mathbf{u}(t) \quad \text{(linearized dynamics)} \tag{9.1.10}$$

$$[\mathbf{z} - \mathbf{h}(\mathbf{x}^*, t)] = \left[\frac{\partial \mathbf{h}}{\partial \mathbf{x}}\right]_{\mathbf{x}=\mathbf{x}^*} \cdot \Delta\mathbf{x} + \mathbf{v}(t) \quad \text{(linearized measurement equation)} \tag{9.1.11}$$

Note that the "measurement" in the linear model is the actual measurement less that predicted by the nominal trajectory in the absence of noise. Also the equivalent $\mathbf{F}$ and $\mathbf{H}$ matrices are obtained by evaluating the partial derivative matrices (Eqs. 9.1.8) along the *nominal* trajectory. We will now look at two examples that illustrate the linearization procedure. In the first example the nonlinearity appears only in the measurement relationship, so it is relatively simple. In the second, nonlinearity occurs in both the measurement and process dynamics, so it is somewhat more involved than the first.

EXAMPLE 9.1

In many electronic navigation systems the basic observable is a noisy measurement of range (distance) from the vehicle to a known location. One such system that has enjoyed wide use in aviation is distance-measuring equipment (DME) (4). We do not need to go into detail here as to how the equipment works. It suffices to say that the airborne equipment transmits a pulse that is returned by the ground station, and then the aircraft equipment interprets the transit time in

terms of distance. In our example we will simplify the geometric situation by assuming that the aircraft and the two DME stations are all in a horizontal plane as shown in Fig. 9.2 (slant range $\approx$ horizontal range). The coordinates of the two DME stations are assumed to be known, and the aircraft coordinates are unknown and to be estimated.

We will look at the aircraft dynamics first. This, in turn, will determine the process state model. To keep things as simple as possible, we will assume a nominal straight-and-level flight condition with constant velocity. The true trajectory will be assumed to be the nominal one plus small perturbations due to random horizontal accelerations, which will be assumed to be white. This leads to random walk in velocity and integrated random walk in position. This is probably unrealistic for long time spans because of the control applied by the pilot (or autopilot). However, this would be a reasonable model for short intervals of time. The basic differential equations of motion in the x and y directions are then

$$\begin{aligned} \ddot{x} &= 0 \quad + u_x \\ \ddot{y} &= \underbrace{0}_{\text{Deterministic forcing function}} + \underbrace{u_y}_{\text{Random forcing function}} \end{aligned} \tag{9.1.12}$$

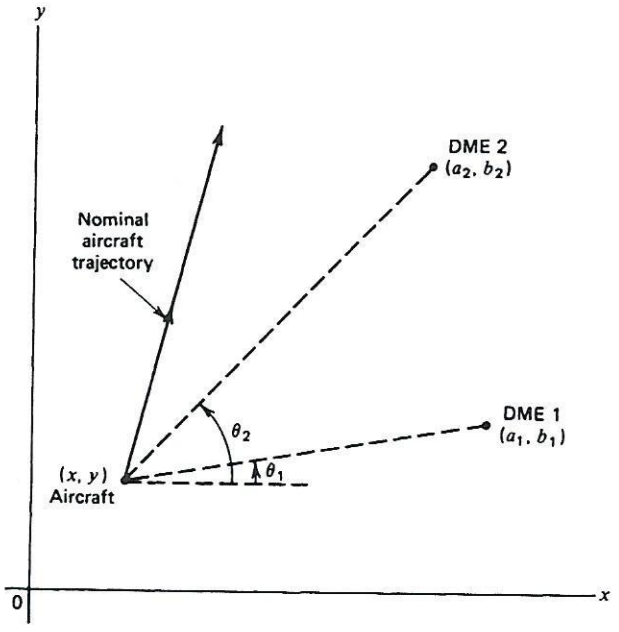


Figure 9.2 Geometry for DME example.

The dynamical equations are seen to be linear in this case, so the differential equations for the incremental quantities are the same as for the total x and y, that is,

$$\Delta\ddot{x} = u_x$$

$$\Delta\ddot{y} = u_y \tag{9.1.13}$$

We now define filter state variables in terms of the incremental positions and velocities:

$$x_1 = \Delta x, \qquad x_2 = \Delta\dot{x}$$

$$x_3 = \Delta y, \qquad x_4 = \Delta\dot{y} \tag{9.1.14}$$

The state equations are then

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} + \begin{bmatrix} 0 \\ u_x \\ 0 \\ u_y \end{bmatrix} \tag{9.1.15}$$

The state variables are driven by the white-noise processes u_x and u_y, so we are assured that the corresponding discrete equations will be in the appropriate form for a Kalman filter.

We now turn to the measurement relationships. We will assume that we have two simultaneous range measurements, one to DME_1 and the other to DME_2. The two measurement equations in terms of the total x and y are then

$$z_1 = \sqrt{(x - a_1)^2 + (y - b_1)^2} + v_1$$

$$z_2 = \sqrt{(x - a_2)^2 + (y - b_2)^2} + v_2 \tag{9.1.16}$$

where v_1 and v_2 are additive white measurement noises. We see immediately that the connection between the observables (z_1 and z_2) and the quantities to be estimated (x and y) is nonlinear. Thus, linearization about the nominal trajectory is in order. We assume that an approximate nominal position is known at the time of the measurement, and that the locations of the two DME stations are known exactly. We now need to form the $\partial\mathbf{h}/\partial\mathbf{x}$ matrix as specified by Eq. (9.1.8). [We note a small notational problem here. The variables x_1, x_2, x_3, and x_4 are used in Eq. (9.1.8) to indicate total state variables, and then the same symbols are used again to indicate incremental state variables as defined by Eqs. (9.1.14). However, the meanings of the symbols are never mixed in any one set of equations, so this should not lead to confusion.] We now note that the x and y position variables are the first and third elements of the state vector. Thus, evaluation of the partial derivatives indicated in Eq. (9.1.8) leads to

$$\frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \begin{bmatrix} \dfrac{(x_1 - a_1)}{\sqrt{(x_1 - a_1)^2 + (x_3 - b_1)^2}} & 0 & \dfrac{(x_3 - b_1)}{\sqrt{(x_1 - a_1)^2 + (x_3 - b_1)^2}} & 0 \\ \dfrac{(x_1 - a_2)}{\sqrt{(x_1 - a_2)^2 + (x_3 - b_2)^2}} & 0 & \dfrac{(x_3 - b_2)}{\sqrt{(x_1 - a_2)^2 + (x_3 - b_2)^2}} & 0 \end{bmatrix} \tag{9.1.17}$$

or

$$\frac{\partial \mathbf{h}}{\partial \mathbf{x}} = \begin{bmatrix} -\cos\theta_1 & 0 & -\sin\theta_1 & 0 \\ -\cos\theta_2 & 0 & -\sin\theta_2 & 0 \end{bmatrix} \tag{9.1.18}$$

Finally, we note that Eq. (9.1.18) can be generalized even further, since the sine and cosine terms are actually direction cosines between the x and y axes and the respective lines of sight to the two DME stations. Therefore, we will write the linearized $\mathbf{H}$ matrix in its final form as

$$\mathbf{H} = \left.\frac{\partial \mathbf{h}}{\partial \mathbf{x}}\right|_{\mathbf{x}=\mathbf{x}^*} = \begin{bmatrix} -\cos\theta_{x1} & 0 & -\cos\theta_{y1} & 0 \\ -\cos\theta_{x2} & 0 & -\cos\theta_{y2} & 0 \end{bmatrix} \tag{9.1.19}$$

where the subscripts on θ indicate the respective axes and lines of sight to the DME stations. Note that $\mathbf{H}$ is evaluated at a point on the *nominal* trajectory. (The true trajectory is not known to the filter.) The nominal aircraft position will change with each step of the recursive process, so the terms of $\mathbf{H}$ are time-variable and must be recomputed with each recursive step. Also, recall from Eq. (9.1.11) that the measurement presented to the linearized filter is the total $\mathbf{z}$ minus the predicted $\mathbf{z}$ based on the nominal position $\mathbf{x}^*$.

Strictly speaking, the linearized filter is always estimating incremental quantities, and then the total quantity is reconstructed by adding the incremental estimate to the nominal part. However, we will see later that when it comes to the actual mechanics of handling the arithmetic on the computer, we can avoid working with incremental quantities if we choose to do so. This is discussed further in the section on the extended Kalman filter. We will now proceed to a second linearization example, where the process dynamics as well as the measurement relationship has to be linearized. ■

EXAMPLE 9.2

This example is taken from Sorenson (5) and is a classic example of linearization of a nonlinear problem. Consider a near-earth space vehicle in a nearly circular orbit. It is desired to estimate the vehicle's position and velocity on the basis of a sequence of angular measurements made with a horizon sensor. With reference to Fig. 9.3, the horizon sensor is capable of measuring:

1. The angle γ between the earth's horizon and the local vertical.
2. The angle α between the local vertical and a known reference line (say, to a celestial object).

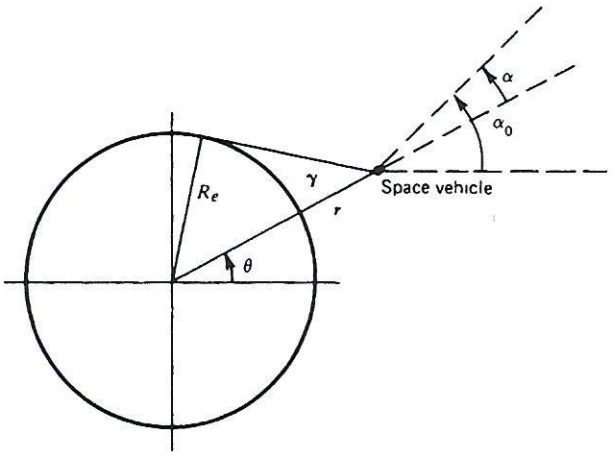


Figure 9.3 Coordinates for space vehicle example.

In the interest of simplicity, we assume all motion and measurements to be within a plane as shown in Fig. 9.3. Thus, the motion of the vehicle can be described with the usual polar coordinates r and θ.

The equations of motion for the space vehicle may be obtained from either Newtonian or Lagrangian mechanics. They are (see Section 2-10, ref. 6):

$$\ddot{r} - r\dot{\theta}^2 + \frac{K}{r^2} = u_r(t) \tag{9.1.20}$$

$$r\ddot{\theta} + 2\dot{r}\dot{\theta} = u_\theta(t) \tag{9.1.21}$$

where K is a constant proportional to the universal gravitational constant, and u_r and u_θ are small random forcing functions in the r and θ directions (due mainly to gravitational anomalies unaccounted for in the K/r^2 term). It can be seen that the constant K must be equal to gR_e^2 if the gravitational forcing function is to match the earth's gravity constant g at the surface. The random forcing functions u_r and u_θ will be assumed to be white. We will look at the linearized process dynamics first and then consider the nonlinear measurement situation later.

The equations of motion, Eqs. (9.1.20) and (9.1.21), are clearly nonlinear, so we must linearize the dynamics if we are to apply Kalman filter methods. We have assumed that random forcing functions u_r and u_θ are small, so the corresponding perturbations from a circular orbit will also be small. By direct substitution into Eqs. (9.1.20) and (9.1.21), it can be verified that

$$r^* = R_0 \quad \text{(a constant radius)} \tag{9.1.22}$$

$$\theta^* = \omega_0 t \quad \left(\omega_0 = \sqrt{\frac{K}{R_0^3}}\right) \tag{9.1.23}$$

will satisfy the differential equations. Thus, this will be the reference trajectory that we linearize about.

We note that we have two second-order differential equations describing the dynamics. Therefore, we must have four state variables in our state model. We choose the usual phase variables as state variables as follows:

$$x_1 = r, \qquad x_2 = \dot{r}$$

$$x_3 = \theta, \qquad x_4 = \dot{\theta} \tag{9.1.24}$$

The nonlinear state equations are then

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = x_1 x_4^2 - \frac{K}{x_1^2} + u_r(t)$$

$$\dot{x}_3 = x_4$$

$$\dot{x}_4 = -\frac{2x_2 x_4}{x_1} + \frac{u_\theta(t)}{x_1} \tag{9.1.25}$$

We must now form the $\partial \mathbf{f}/\partial \mathbf{x}$ matrix indicated in Eq. (9.1.8) to get the linearized $\mathbf{F}$ matrix.

$$\frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \left(x_4^2 + \dfrac{2K}{x_1^3}\right) & 0 & 0 & 2x_1 x_4 \\ 0 & 0 & 0 & 1 \\ \dfrac{2x_2 x_4}{x_1^2} & \dfrac{-2x_4}{x_1} & 0 & \dfrac{-2x_2}{x_1} \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 & 0 & 0 \\ \left(\dot{\theta}^2 + \dfrac{2K}{r^3}\right) & 0 & 0 & 2r\dot{\theta} \\ 0 & 0 & 0 & 1 \\ \dfrac{2\dot{r}\dot{\theta}}{r^2} & \dfrac{-2\dot{\theta}}{r} & 0 & \dfrac{-2\dot{r}}{r} \end{bmatrix} \tag{9.1.26}$$

Next, we evaluate $\partial \mathbf{f}/\partial \mathbf{x}$ along the reference trajectory.

$$\left.\frac{\partial \mathbf{f}}{\partial \mathbf{x}}\right|_{\substack{r=R_0 \\ \theta=\omega_0 t}} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 3\omega_0^2 & 0 & 0 & 2R_0\omega_0 \\ 0 & 0 & 0 & 1 \\ 0 & \dfrac{-2\omega_0}{R_0} & 0 & 0 \end{bmatrix} \tag{9.1.27}$$

Equation (9.1.27) then defines the **F** matrix that characterizes the linearized dynamics. Note that in the linear equations, Δr, $\dot{\Delta} r$, $\Delta\theta$, and $\dot{\Delta}\theta$ become the four state variables.

We now turn to the measurement model. The idealized (no noise) relationships are given by

$$\begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} \gamma \\ \alpha \end{bmatrix} = \begin{bmatrix} \sin^{-1}\left(\dfrac{R_e}{r}\right) \\ \alpha_0 - \theta \end{bmatrix} \tag{9.1.28}$$

We next replace r with x_1 and θ with x_3, and then perform the partial derivatives indicated by Eq. (9.1.8). The result is

$$\left[\frac{\partial \mathbf{h}}{\partial \mathbf{x}}\right] = \begin{bmatrix} -\dfrac{R_e}{r\sqrt{r^2 - R_e^2}} & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \tag{9.1.29}$$

Finally, we evaluate $\partial \mathbf{h}/\partial \mathbf{x}$ along the reference trajectory

$$\left.\frac{\partial \mathbf{h}}{\partial \mathbf{x}}\right|_{\substack{r=R_0 \\ \theta=\omega_0 t}} = \begin{bmatrix} -\dfrac{R_e}{R_0\sqrt{R_0^2 - R_e^2}} & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{bmatrix} \tag{9.1.30}$$

This then becomes the linearized **H** matrix of the Kalman filter. The linearized model is now complete with the determination of the **F** and **H** matrices. Before we leave this example, though, it is worth noting that the forcing function $u_\theta(t)$ must be scaled by $1/R_0$ in the linear model because of the $1/x_1$ factor in Eq. (9.1.25). ■

The Extended Kalman Filter

The extended Kalman filter is similar to a linearized Kalman filter except that the linearization takes place about the filter's estimated trajectory, as shown in

Fig. 9.4, rather than a precomputed nominal trajectory. That is, the partial derivatives of Eq. (9.1.8) are evaluated along a trajectory that has been updated with the filter's estimates; these, in turn, depend on the measurements, so the filter gain sequence will depend on the sample measurement sequence realized on a particular run of the experiment. Thus, the gain sequence is not predetermined by the process model assumptions as in the usual Kalman filter.

A general analysis of the extended Kalman filter is difficult because of the feedback of the measurement sequence into the process model. However, qualitatively it would seem to make sense to update the trajectory that is used for the linearization—after all, why use the old trajectory when a better one is available? The flaw in this argument is this: The "better" trajectory is only better in a *statistical* sense. There is a chance (and maybe a good one) that the updated trajectory will be poorer than the nominal one. In that event, the estimates may be poorer; this, in turn, leads to further error in the trajectory, which causes further errors in the estimates, and so forth and so forth, leading to eventual divergence of the filter. The net result is that the extended Kalman filter is a somewhat riskier filter than the regular linearized filter, especially in situations where the initial uncertainty and measurement errors are large. It may be better *on the average* than the linearized filter, but it is also more likely to diverge in unusual situations.

Both the regular linearized Kalman filter and the extended Kalman filter have been used in a variety of applications. Each has its advantages and disadvantages, and no general statement can be made as to which is best because it depends on the particular situation at hand. Aided inertial navigation systems serve as good examples of both methods of linearization, and this is discussed further in Chapter 10.

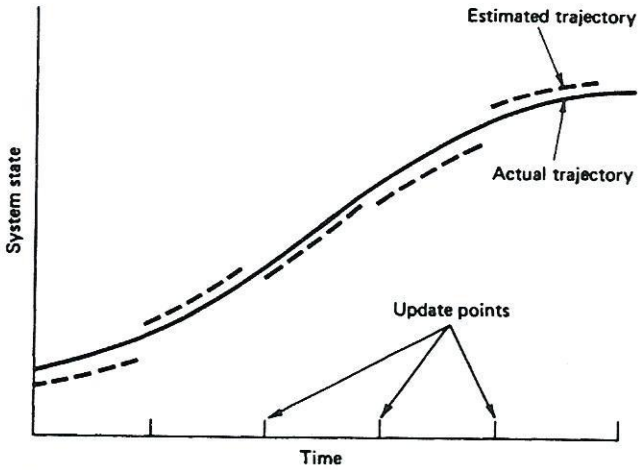


Figure 9.4 Reference and actual trajectories for an extended Kalman filter.

Keeping Track of Total Estimates in an Extended Kalman Filter

It should be remembered that the basic state variables in a linearized Kalman filter are incremental quantities, and not the total quantities such as position, velocity, and so forth. However, in an extended Kalman filter it is usually more convenient to keep track of the total estimates rather than the incremental ones, so we will now proceed to show how this is done and why it is valid to do so.

We begin with the basic linearized measurement equation, Eq. (9.1.11)

$$\mathbf{z} - \mathbf{h}(\mathbf{x}^*) = \mathbf{H}\Delta\mathbf{x} + \mathbf{v} \tag{9.1.31}$$

Note that when working with incremental state variables, the measurement presented to the Kalman filter is $[\mathbf{z} - \mathbf{h}(\mathbf{x}^*)]$ rather than the total measurement $\mathbf{z}$. Next, consider the incremental estimate update equation at time t_k.

$$\Delta\hat{\mathbf{x}}_k = \Delta\hat{\mathbf{x}}_k^- + \mathbf{K}_k \underbrace{[\mathbf{z}_k - \mathbf{h}(\mathbf{x}_k^*) - \mathbf{H}_k\Delta\hat{\mathbf{x}}_k^-]}_{\text{Inc. meas.}} \tag{9.1.32}$$

Now, in forming the measurement residual in Eq. (9.1.32), suppose we associate the $\mathbf{h}(\mathbf{x}_k^*)$ term with $\mathbf{H}_k\Delta\hat{\mathbf{x}}_k^-$ rather than $\mathbf{z}_k$. This measurement residual can then be written as

$$\text{Measurement residual} = (\mathbf{z}_k - \hat{\mathbf{z}}_k^-) \tag{9.1.33}$$

because the predictive estimate of the measurement is just the sum of $\mathbf{h}(\mathbf{x}_k^*)$ and $\mathbf{H}_k\Delta\hat{\mathbf{x}}_k^-$. Note that the measurement residual as given by Eq. (9.1.33) is formed exactly as would be done in an extended Kalman filter, that is, it is the noisy measurement minus the predictive measurement based on the corrected trajectory rather than the nominal one.

We now return to the update equation, Eq. (9.1.32), and add $\mathbf{x}_k^*$ to both sides of the equation:

$$\underbrace{\mathbf{x}_k^* + \Delta\hat{\mathbf{x}}_k}_{\hat{\mathbf{x}}_k} = \underbrace{\mathbf{x}_k^* + \Delta\hat{\mathbf{x}}_k^-}_{\hat{\mathbf{x}}_k^-} + \mathbf{K}_k(\mathbf{z}_k - \hat{\mathbf{z}}_k^-) \tag{9.1.34}$$

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_k^- + \mathbf{K}_k(\mathbf{z}_k - \hat{\mathbf{z}}_k^-) \tag{9.1.35}$$

Equation (9.1.35) is, of course, the familiar linear estimate update equation written in terms of *total* rather than incremental quantities. It simply says that we correct the a priori estimate by adding the measurement residual appropriately weighted by the Kalman gain $\mathbf{K}_k$. Note that after the update is made in the extended Kalman filter, the incremental $\Delta\hat{\mathbf{x}}_k$ is reduced to zero. Its projection to

the next step is then trivial. The only nontrivial projection is to project $\hat{\mathbf{x}}_k$ (which has become the reference $\mathbf{x}$ at t_k) to $\hat{\mathbf{x}}_{k+1}^-$. This must be done through the nonlinear dynamics as dictated by Eq. (9.1.1). That is,

$$\hat{\mathbf{x}}_{k+1}^- = \left\{ \begin{array}{l} \text{Solution of the nonlinear differential equation} \\ \dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}_d, t) \text{ at } t = t_{k+1}, \text{ subject to the} \\ \text{initial condition } \mathbf{x} = \hat{\mathbf{x}}_k \text{ at } t_k \end{array} \right\} \tag{9.1.36}$$

Note that the additive white-noise forcing function $\mathbf{u}(t)$ is zero in the projection step, but the deterministic $\mathbf{u}_d$ is included in the $\mathbf{f}$ function. Once $\hat{\mathbf{x}}_{k+1}^-$ is determined, the predictive measurement $\hat{\mathbf{z}}_{k+1}^-$ can be formed as $\mathbf{h}(\hat{\mathbf{x}}_{k+1}^-)$, and the measurement residual at t_{k+1} is formed as the difference $(\mathbf{z}_{k+1} - \hat{\mathbf{z}}_{k+1}^-)$. The filter is then ready to go through another recursive loop.

For completeness, we repeat the familiar error covariance update and projection equations:

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^- \tag{9.1.37}$$

$$\mathbf{P}_{k+1}^- = \boldsymbol{\phi}_k\mathbf{P}_k\boldsymbol{\phi}_k^T + \mathbf{Q}_k \tag{9.1.38}$$

where $\boldsymbol{\phi}_k$, $\mathbf{H}_k$, and $\mathbf{Q}_k$ come from the linearized model. Equations (9.1.37) and (9.1.38) and the gain equation (which is the same as in the linear Kalman filter) should serve as a reminder that the extended Kalman filter is still working in the world of linear dynamics, even though it keeps track of total estimates rather than incremental ones.

Getting the Extended Kalman Filter Started

It was mentioned previously that the extended Kalman filter can diverge if the reference about which the linearization takes place is poor. The most common situation of this type occurs at the initial starting point of the recursive process. Frequently, the a priori information about the true state of the system is poor. This causes a large error in $\hat{\mathbf{x}}_0^-$ and forces $\mathbf{P}_0^-$ to be large. Thus, two problems can arise in getting the extended filter started:

1. A very large $\mathbf{P}_0^-$ combined with low-noise measurements at the first step will cause the $\mathbf{P}$ matrix to "jump" from a very large value to a small value in one step. In principle, this is permissible. However, this can lead to numerical problems due to roundoff. A non-positive-definite $\mathbf{P}$ matrix at any point in the recursive process usually leads to divergence.
2. The initial $\hat{\mathbf{x}}_0^-$ is presumably the best estimate of $\mathbf{x}$ prior to receiving any measurement information, and thus it is used as the reference for linearization. If the error in $\hat{\mathbf{x}}_0^-$ is large, the first-order approximation

used in the linearization will be poor, and divergence may occur, even with perfect arithmetic.

With respect to problem 1, the filter designer should be especially careful to use all the usual numerical precautions to preserve the symmetry and positive definiteness of the **P** matrix on the first step. In some cases, simply using the symmetric form of the **P**-update equation is sufficient to ward off divergence. This form, Eq. (5.5.11), is repeated here for convenience (sometimes called the Joseph form; see ref. 7):

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)\mathbf{P}_k^-(\mathbf{I} - \mathbf{K}_k\mathbf{H}_k)^T + \mathbf{K}_k\mathbf{R}_k\mathbf{K}_k^T \tag{9.1.39}$$

Another way of mitigating the numerical problem is to let $\mathbf{P}_0^-$ be considerably smaller than would normally be dictated by the true a priori uncertainty in $\hat{\mathbf{x}}_0^-$. This will cause suboptimal operation for the first few steps, but this is better than divergence! A similar result can be accomplished by letting $\mathbf{R}_k$ be abnormally large for the first few steps. There is no one single cure for all numerical problems. Each case must be considered on its own merits.

Problem 2 is more subtle than problem 1. Even with perfect arithmetic, poor linearization can cause a poor $\hat{\mathbf{x}}_0^-$ to be updated into an even poorer a posteriori estimate, which in turn gets projected on ahead, and so forth. Various "fixes" have been suggested for the poor-linearization problem, and it is difficult to generalize about them (7, 8, 9, 10). All are ad hoc procedures. This should come as no surprise, because the extended Kalman filter is, itself, an ad hoc procedure. One remedy that works quite well when the information contained in $\mathbf{z}_0$ is sufficient to determine $\mathbf{x}$ algebraically is to use $\mathbf{z}_0$ to solve for $\mathbf{x}$, just as if there were no measurement error. This is usually done with some tried-and-true numerical algorithm such as the Newton–Raphson method of solving algebraic equations (11). It is hoped this will yield a better estimate of $\mathbf{x}$ than the original coarse $\hat{\mathbf{x}}_0^-$. The filter can then be linearized about the new estimate (and a smaller $\mathbf{P}_0^-$ than the original $\mathbf{P}_0^-$ can be used), and the filter is then run as usual beginning with $\mathbf{z}_0$ and with proper accounting for the measurement noise. Another ad hoc procedure that has been used is to let the filter itself iterate on the estimate at the first step. The procedure is fairly obvious. The linearized filter parameters that depend on the reference $\mathbf{x}$ are simply relinearized with each iteration until convergence is reached within some predetermined tolerance. $\mathbf{P}_0^-$ may be held fixed during the iteration, but this need not be the case. Also, if $\mathbf{x}$ is not observable on the basis of just one measurement, iteration may also have to be carried out at a few subsequent steps in order to converge on good estimates of all the elements of $\mathbf{x}$. There is no guarantee that iteration will work in all cases, but it is worth trying.

Before leaving the subject of getting the filter started, it should be noted that neither the algebraic solution nor the iteration remedies just mentioned play any role in the basic "filtering" process. Their sole purpose is simply to provide a good reference for linearization, so that the linearized Kalman filter can do its job of optimal estimation.